Blood Group Detection through Finger Print

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Abstract: Fingerprints are essential for blood group identification. to diagnose a patient without waiting for traditional medical formalities and to anticipate the patient's blood type in emergency situations. concentrating on obtaining accurate and good results for everyday medical use in hospitals. It will be more beneficial to avoid time-consuming techniques throughout the critical time of medication. In this study, the only correlation between gender and finger print patterns was that females were more likely than males to have loops and arches, and males were more likely than females to have whorls. For a long time, fingerprint identification has been considered one of the most reliable ways to identify someone, especially in court. Fingerprints are believable because, with the exception of severe skin injuries, the patterns we create while still in the womb don't change throughout our lives. The fact that no two fingerprints are alike is even more amazing. Because each fingerprint has a unique pattern, even if they are from the same person, they are reliable pieces of evidence in court. The purpose of our study was to determine whether blood group, gender, and fingerprint patterns are related in any way. We looked at 200 people who lived in Hubli-Dharwad, Karnataka, India; half of them were women and the other half were men, and they ranged in age. The fingerprints were divided into three groups: arches, whorls, and loops. Our findings showed that arches were the least common motif and loops were by far the most common. Additionally, we found that loops happened in people with all blood types, including Rh-negative and Rh-positive. However, it seemed that those with O-negative blood had more whorls. In terms of gender, we found that men's fingerprints had more whorls, while women's had more loops and arches. This suggests that blood type and gender may be linked to fingerprint patterns, and in certain cases, we may even be able to predict a person's blood type or gender just by looking at their fingerprints.

Keywords - Blood Group, Machine Learning, Finger print, Medical Field.

I. INTRODUCTION

Generally fingerprints are used to give an unique identity for individuals to generate Public IDs but in this case fingerprints can actually help to predict accurate blood group and avoid latency to the patients who are in crucial stage can get medication as priority. The patterns in the fingerprint can be differentiated based on the Proteins and Antigens presence in individual blood type on fingertips. The goal focuses on giving an Accurate reading of blood group for Efficient and easy blood group detection to help Medical Field. The skin covering the human hand's anterior surface and the foot's planter surface differs from the skin covering the rest of the body in terms of texture and appearance. The palmar and plantar surfaces of this skin are constantly wrinkled and have tiny, narrow ridges called friction ridges

The impression of all the friction ridges on a finger is called a fingerprint. Dermal carvings, also known as finger prints, first appear on human fingers, palms, soles, and toes between weeks 12 and 16 of embryonic development. By week 14, or roughly the sixth month of pregnancy, their formation is complete. Thus, The evolution of Fingerprints Occur.

II. LITERATURE SURVEY

Blood Group Detection Using Fingerprint Images Authors: Vidya Waykule, Sakshi Amrutkar, Om Jadhav, Vardhman Jain, Titiksha Jawale.

A noteworthy development in biometric-based medical diagnostics is the blood group prediction project, which made use of fingerprint analysis and advanced machine learning techniques.
This technology uses powerful classification algorithms in conjunction with fingerprint data to attempt to deliver quick, non-invasive, and accurate blood group identification. For several applications in forensics, emergency care, and medical record administration, this offers a practical option. This approach has a lot of potential because of its advantages, which include high accuracy, efficiency, and usability. However, there are still problems, particularly with regard to the unpredictable nature of datasets, the need for enormous volumes of data, and potential technical limitations in the feature extraction process for fingerprint images.

Analysis of Fingerprint Patterns in Connection to Gender and Blood Group: A Statistical Review Karnataka Science College, Department of Criminology and Forensic Science, Dharwad, Karnataka, India; Desai Bhavana1, Jaiswal Ruchi2, Tiwari Prakash2, and Kalyan J.L.1 1 2Dr. Harising Gour Vishwavidyalaya, Department of Criminology and Forensic Science, Sagar, MP, India.

This study provides us with an overview of the occurrences of specific pattern types that are associated with blood group.

- While Arches are rather uncommon, loops seem to be the most common fingerprint type.
- O negative and AB negative blood types are the rarest in the study, whereas B positive blood is the most common.

- The discovery that people with blood type O negative are more likely to have whorls than people with other Rh-negative blood types is consistent with earlier studies by Dr. Prateek Rastogi and Ms. Keerti R. Pillai.

People with blood types B and O showed a tendency to have more Loops and Arches compared to other groups, possibly indicating a genetic connection between blood type and fingerprint pattern.

- The greatest number of whorls was seen in both B and O positive blood types, which may indicate a closer relationship between these blood types and whorl patterns.

- There may be a distinct pattern for blood type AB, as evidenced by the uncommon frequency of Arches in this population.

- Fingerprint patterns appear to differ by gender, with males having a higher frequency of Whorls and females having more Loops and Arches. However, we just want to identify the blood type, not the gender.

- Compared to blood groups A and AB, Loops were significantly more common in blood groups B and O.

Relationship between pattern of fingerprints and blood groups
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This study investigates the connection between gender, blood types, and the distribution of dermatoglyphic characteristics, or fingerprints.
450 individuals in all were chosen at random from Koya University's faculties. Of these, 226 were women and 224 were men, and the bulk of participants had blood group O.
In particular, 16 (3.55%) were Rh-negative O and 165 (36.66%) were Rh-positive O. Following this, 124 people (27.55%) had blood group A, 9 people (2%), and 87 people (19.33%) had blood group A. 34 (7.55%) AB-positive, 3 (0.66%) AB-negative, 12 (2.66%) B-negative, and B-positive.
Forty (8.8%) participants were Rh-negative, whereas the majority (91.1%, or 410 people) were Rh-positive. Both male and female individuals had the highest frequency of blood group O, followed by groups A, B, and AB.
These results are consistent with those of Rastogi and Pillai (2010) and Bharadwaj and Ewoldt (2015).
Desai et al. (2013) and Mehta and Mehta (2011), on the other hand, noted that blood type B was the most prevalent, followed by O and A.

Blood group detection using fingerprints: developments and technologies First K Bangalore, Karnataka.
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By providing a non-invasive, effective, and convenient substitute for conventional blood type techniques, fingerprint-based blood group identification is a new and creative technique that has the potential to completely transform medical diagnostics. For this technology to reach its full potential, several important obstacles must be overcome. Among these are sparse, skewed datasets, a dearth of empirical validation, uneven approaches, and challenges with feature extraction and population-wide generalization.
Future research should focus on:
Developing larger and more diverse datasets.
Standardizing methodologies to ensure consistency and reliability.
Testing these systems in real-world environments to ensure scalability and practicality.
The use of advanced machine learning techniques, such as deep learning, and fostering interdisciplinary collaboration can lead to more accurate, robust, and innovative solutions for fingerprint-based blood group detection systems.

By overcoming these challenges, researchers can move closer to creating a versatile and reliable tool for blood group determination that is cost-effective, widely accessible, and capable of improving global healthcare diagnostics.

III. Hardware Specifications

Fingerprint Scanner: To determine ridge count, minutiae points, and texture-related features that are important for blood group classification, it is essential to obtain fingerprint images in critical resolution.

IV. Software Specifications

OpenCV: Essential for the creation of fingerprint images, preprocessing procedures such as segmentation and enhancement techniques are required for analysis purposes, such as the extraction of minutiae points and ridge counts, which are followed by the extraction of regions of interest (ROI).

A popular data management tool for feature extraction is NumPy.

Pandas: Using feature sets to expedite data retrieval for machine learning applications, Pandas is crucial for effectively managing and organizing data.

Mahotas: The Gray Level Cooccurrence Matrix, or GLCM, is used in image processing techniques such as transformations.

Matplotlib and Seaborn: These tools allow you to visualize the feature value distributions, the significance of each feature, and the relationships between the characteristics that have been extracted.

V. Ease of Use

A. *Types of Blood Groups*

Blood groups come in four varieties: A, B, AB, and O. One type of protein in blood that determines whether it is present in a positive or negative state is called the Rh Factor.

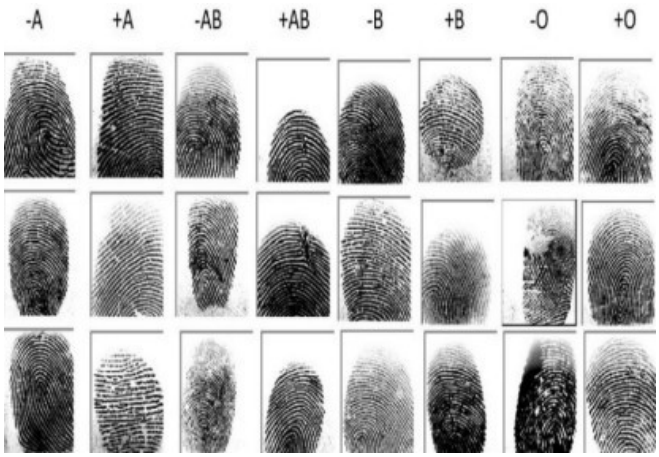


Fig.1: Collection of Finger Prints based on Blood Group

B. Types of Finger Prints

There are total of 4 Main Finger Print Types:

i. Arches

ii. Loops

iii.Whorls

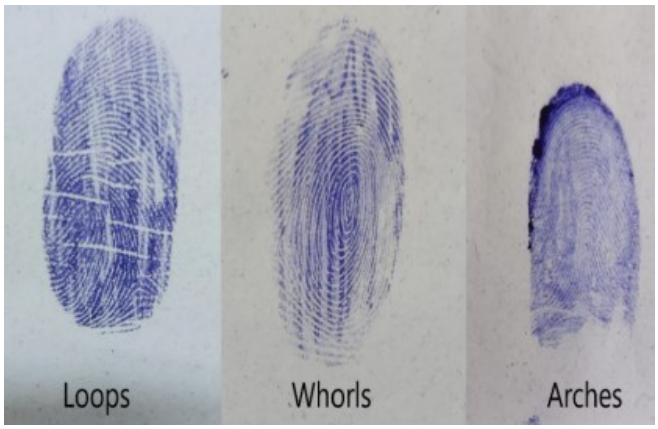
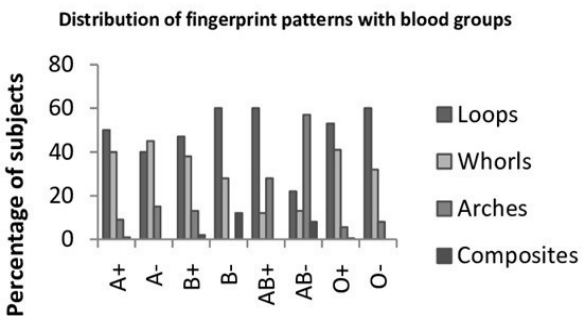


Fig 2: Different Types of Finger Prints Patterns

Additionally, there is a different kind of fingerprint known as accidental. It only occurs in 3-5% of the world's population.

VI. Finding a Proper Dataset

A. Observations



In order to obtain correct sample findings, we require a well-trained dataset. more copies for a certain blood group.

Each sort of fingerprint has a different structure. By using a fingerprint as an example, we can observe that it is a type of real fingerprint. When a few finger prints are compared, they are discovered to be comparable to the same kind in the same blood group.

B. Analysis of Fingerprint

The machine learning model is trained using a particular fingerprint sample in order to identify the blood group from the fingerprint. The model takes the fingerprint pattern as input data and applies statistical correlation techniques between blood groups and fingerprint types. The model then returns the blood group that was determined as an output.

C. Results and Discussion

The findings of these studies show a statistically significant correlation between the matching blood groups and fingerprint patterns. The following findings are suggested by this study:

- Blood group O predominantly exhibit loop patterns.
- Blood group A exhibits a mix of loops and arches, with loops being dominant in most samples.
- Blood group B exhibit domination of whorls.
- Blood group AB exhibits combination of loops and whorls.

With the help of this correlation ,it can be leverged for predictive analysis of blood group.

D. Some Common Mistakes

Due to the large size of the human population and the numerous possibilities for fingerprints, the error rate can be unpredictable. This will be prevented by carefully examining the data and the dataset that is provided in order to lower the error rate.

Although fingerprint-based methods for blood group detection are a novel technology, they are still in the development stage and are not yet commercially available. This approach may contain a variety of mistakes, including:

1. False Negative and False Positive Results

a. Because some people with different blood groups have fingerprints that are similar, the system can provide a false blood type signal.

b. Inaccurate classification may result in mismatched transfusions.

2. Environmental Factors

a. Interference may arise from dirt, dampness, or damage to the fingerprint's surface.

b. Low-quality fingerprint scanning can result in erroneous readings.

3. Technical Limitations

a. Current blood group identification methods based on fingerprints rely on AI or machine learning algorithms that were trained on tiny datasets; their generalizability cannot be ensured.

b. The resolution of fingerprint sensors might not be high enough to pick up micro-level details required for correct classification.

4. Biological Variability

a. The idea that the fingerprint patterns match powerfully

with blood groups remains to be researched, and variations in individuals can cause errors.

b. Genetics, ethnicity, and ailments may affect fingerprint patterns, making accuracy poor.

5. Algorithmic Errors

a. Biased training data causes classification model faults .

b. Overfitting to specific fingerprint traits, which reduces accuracy in real-world applications

6. Lack of Standardization

a. Inconsistent results may be provided by different devices and algorithms.

b. A generally recognized standard for fingerprint-based blood group detection does not yet exist.

Due to their great accuracy and dependability, traditional blood testing methods (such as agglutination tests) remain the gold standard today. Although fingerprint detection can be useful for rapid screening, more research is necessary before it can be used in therapeutic settings.

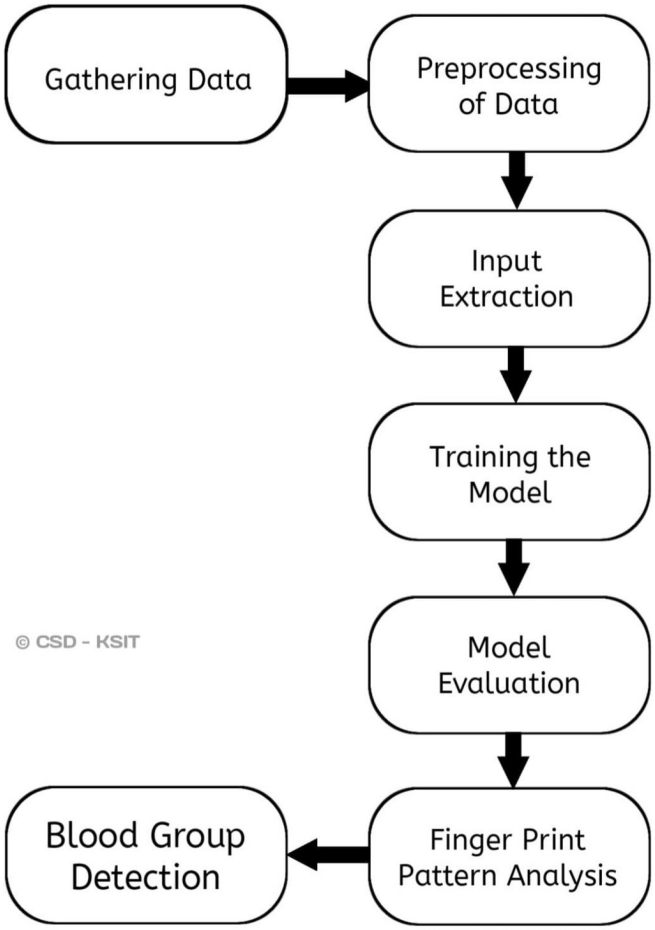


Fig 3. Blood Group detection using Finger Print Pattern Framework

VII. METHODOLOGY

1. Identification of Problem:

The traditional method of blood detection is time consuming, painful, unhygienic and takes more time for analyzing the results. The process of determining the Blood Group is quite complex, So to replace this traditional method with a convenient detection of the group.

2. Data Collection:

We gathered as many fingerprint pattern samples as possible from each blood group and categorized them using a trained dataset. In order to train the machine learning model, we collected a provisional data set of approximately 6000 fingerprint patterns.

An essential step in a study that uses fingerprints to identify blood types is data collection. To train and test the detection system, biometric fingerprint data and related blood group data must be gathered. The crucial steps in the data collection process are as follows:

Biometric Fingerprint Data Collection

Collect fingerprint images using a high-resolution fingerprint scanner. Collect multiple samples per individual to compensate for fingerprint pattern variations.

Store fingerprint data in a structured database with proper labeling.

Blood Group Verification, Enquire the participants by asking their Blood Group in Prior to get a Clarification.

Store blood group information securely along with fingerprint images. Extract important features from fingerprint patterns (e.g., ridges etc). Normalize data to maintain uniformity among all collected samples.

Once data has been gathered and processed, it can be employed to train blood group predicting machine learning models from fingerprint patterns.

3. Preprocessing of Data:

Before training a model and feature extraction we have to process the data which include:

I. Image Enhancement: Adjusting contrast and reduce noise to improve fingerprint image clarity.

II. Normalization: Data is standardized to reduce variation in input due to image quality and scanning conditions.

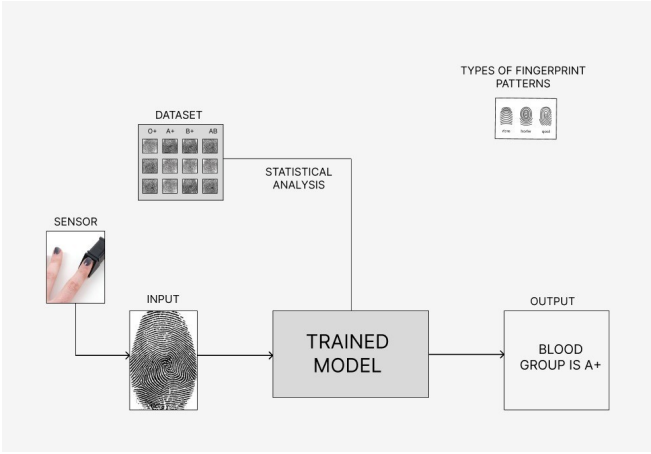


Fig. 4: Work flow of Blood Detection through Patterns

Preprocessing the images to detect blood groups from fingerprints includes some crucial steps to make the images clear and ready to be analyzed. To begin, fingerprint images

are taken and de-noised by removing any noise with the help of filters to make the details prominent. The photographs are all downsized to the same size and the contrast and brightness are adjusted so that differences in light between them do not affect the analysis in order to prevent any inconsistencies. The next stage is to focus on the central piece of the fingerprint, which is the most important part. To highlight the valleys and ridges, this area is singled out and the image is converted to black and white. A technique called ridge thinning is used to make the ridges more visible, simplifying the structure and allowing us to easily spot crucial elements like splits and ridge termination. The direction and frequency of the ridges are then calculated to process the fingerprint pattern as a whole.Finally, techniques like resizing and rotating the photos are used to create more variances in the data in order to strengthen the model. These procedures make it possible to prepare the fingerprint images so that the model can efficiently learn from them and forecast blood groups with accuracy.

4. Feature Extraction:

After the Pre processing of data, minute details like Point, Ridge Count, and fingerprint patterns that is(loops worls, arches or combination of these patterns) are extracted and analyzed to determine blood group.

Pre-processing fingerprint photos to make them clear and noise-free is the first stage in identifying blood types from fingerprints. We extract important fingerprint information, like minutiae points—the minute elements, like splits and ridge endings, that set one fingerprint apart from another—after the image quality has been adjusted. Using methods like Gabor filters, we also look at the ridges' structure and direction. In addition, we examine the texture of the fingerprint with techniques such as the Gray-Level Co-occurrence Matrix (GLCM) and find more general features such as the core and delta points that inform us of the overall type of the fingerprint (e.g., whorl, loop, arch). After obtaining each of these attributes, we reduce their quantity to improve the analysis's effectiveness. In the end, we feed these attributes into a classification algorithm that can determine the blood group from the unique fingerprint patterns, such as Support Vector Machines (SVM) or Neural Networks (ANN).

5. Training the Model:

A predictor models developed using machine leA machine learning-based predictor model that has been trained to identify blood groups. For this project, CNN is the most common machine learning method that could serve as the basis for this model.

Pre-processing fingerprint photos to make sure they are clear and noise-free is the first stage in identifying blood types from fingerprints. Following picture quality optimization, we extract crucial information from the fingerprint, including minutiae points—tiny characteristics like splits and ridge endings that set one fingerprint apart from another. Using Gabor filters, we additionally examine the ridges' orientation and direction. In addition, we evaluate the texture of the fingerprint via techniques such as the Gray-Level Co-occurrence Matrix (GLCM) and detect wider features such as the core and delta points informing us of the overall type of fingerprint (e.g., whorl, loop, arch).

To improve the analysis's efficiency, we restrict the quantity of these qualities after gathering them all. Finally, these characteristics are fed into a classification engine that uses the unique fingerprint patterns to determine the blood group.

Conclusion and Future reference

The viability of fingerprint pattern analysis as a distinct, original, creative, practical, and non-invasive method for blood group identification is demonstrated by this study. Although the results from the current model are notable and encouraging, we may increase the project's efficiency and accuracy as well as streamline the blood detection procedure by utilizing cutting-edge machine learning algorithms and approaches. If this model or concept can provide findings that are 100% accurate, it can be used to the entire medical profession and save money and time compared to the conventional blood group detection approach. To improve predictive skills, other study might examine genetic markers and incorporate deep learning models. We may also improve model performance by refining data preparation methods like noise reduction and ridge feature segmentation. This invention holds great promise for the fields of medical diagnostics, donor-recipient matching in transfusion medicine, and forensic investigations.

Individuality's ridge pattern is a permanent aspect of personal identity that is unchanged from birth. This study presents a method for personal identification systems based on the comparison of specific fingerprint feature patterns. Information from fingerprints is also used to determine blood group. Ridge frequency is recorded during fingerprint matching, and a Gabor filter is used for the same purpose to extract spatial properties. Thus, blood types can be ascertained using fingerprint analysis.

According to the study, arches are the least common fingerprint pattern, while loops are the most common. O- and AB- are the least common blood groups, whereas B+ was the most common. Interestingly, compared to other Rh-negative blood types, those with the O-blood type exhibited more whorls in their fingerprints. While blood types B+ and O+ displayed the greatest number of whorls, blood groups B and O displayed more arches and loops. In addition, men displayed more whorls, while females displayed more loops and arches.

Human fingerprints' intricate ridge patterns are incredibly accurate identification markers since they remain constant throughout an individual's life. In this article, a sophisticated identity identification approach based on the study of distinctive fingerprint traits is discussed.
In order to ensure reliable identification, it is recommended that distinct spatial features be extracted from fingerprint images and compared to a reference database.

Fingerprint analysis has been utilized in biological research, particularly in determining a person's blood group, in addition to the traditional identity verification process. This study investigates the connection between fingerprint ridge patterns and blood type classification.
In order to recognize a fingerprint, ridge frequency must be measured and intricate image processing techniques, such as Gabor filters, must be used to extract textural and spatial characteristics.

Scientists want to establish a potential link between dermatoglyphic patterns and blood type classification by analyzing these fingerprint-based characteristics.

This study illustrates fingerprint analysis's dual usefulness as a non-invasive method for predicting blood group information and as a powerful biometric identification tool. The findings of this study could lead to new uses in both medical diagnostics and forensic science.

Types of Finger Print	Blood group A		Blood group B		Blood group AB		Blood group O	
	Rh +	Rh -	Rh +	Rh -	Rh+	Rh -	Rh+	Rh-
Loops	335 63.2 %	24 80%	38 62.79%	23 76.67%	62 68.8%	0	385 55.79%	9(45%)
Whorls	168 31.7 %	3 10%	188 30.82%	6 20%	27 30 %	0	246 35.65%	11(55%)
Arches	27 5.09%	3 10%	39 6.39%	1 3.33%	1 1.11 %	0	59 8.6%	0
Total	530	30	610	30	90	0	690	20

Fig. 5: Overall Analysis of Blood groups through Finger Prints

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